

Lab 2

Access Modifiers

Access Modifiers or Access specifiers define how the members (attributes and methods) of a class can be accessed.

In C++, there are three access specifiers:

- **public** - members are accessible from outside the class.
- **private** - members cannot be accessed (or viewed) from outside the class.
- **protected** - members cannot be accessed from outside the class, however, they can be accessed in inherited classes.

Example 1:

```
#include <iostream>

using namespace std;

class Line {
    public:
        double length;
        void setLength( double len );
        double getLength( void );
};

// Member functions definitions
double Line::getLength(void) {
    return length ;
}

void Line::setLength( double len) {
    length = len;
}

// Main function for the program
int main() {
    Line line;

    // set line length
```

```
    line.setLength(6.0);  
    cout << "Length of line : " << line.getLength()  
<<endl;  
  
    // set line length without member function  
    line.length = 10.0; // OK: because length is public  
    cout << "Length of line : " << line.length <<endl;  
  
    return 0;  
}
```

Example 2:

```
#include <iostream>  
  
using namespace std;  
  
class Box {  
    public:  
        double length;  
        void setWidth( double wid );  
        double getWidth( void );  
  
    private:  
        double width;  
};  
  
// Member functions definitions  
double Box::getWidth(void) {  
    return width ;  
}  
  
void Box::setWidth( double wid ) {  
    width = wid;  
}  
  
// Main function for the program  
int main() {  
    Box box;  
  
    // set box length without member function  
    box.length = 10.0; // OK: because length is public  
    cout << "Length of box : " << box.length <<endl;  
  
    // set box width without member function
```

Object Oriented Programming

```
// box.width = 10.0; // Error: because width is
private
box.setWidth(10.0); // Use member function to set
it.
cout << "Width of box : " << box.getWidth() << endl;

return 0;
}
```